

10439**C2 Power system operation & control****PS1 Create operational resilience to extreme/unpredictable events****Verification of a 3-Dimensional Geoelectric Field Model for Geomagnetic Disturbance and Geomagnetically Induced Current Studies**

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SUMMARY

During geomagnetic disturbances (GMD), quasi-direct (dc) currents are induced along transmission lines and through transformer windings to a path to ground, i.e., the grounded-wye transformer neutral. The North American Electric Reliability Corporation (NERC) Standard TPL-007 mandates Transmission Owners to perform Steady State analysis to simulate GMD events and evaluate these geomagnetically induced current (GIC) levels and impacts.

Calculating GIC in a power flow software tool requires a geoelectric field (E-field) input. Currently the process called for in TPL-007 uses a static E-field, i.e., 8 V/km for the benchmark event and 12 V/km for a supplemental event over a localized area, which is then scaled based on the latitude and also scaled based on a simple 1-dimensional (1-D) Earth conductivity model specified for the region where the power grid is located. This static geoelectric field has fixed directional orientation across the geographic region and is input to the model to predict GIC in each transformer neutral.

It is known, however, that the geoelectric field is non-uniform in time and space. In response to the need for a more accurate assessment of local GIC hazard, the United States Geological Survey (USGS) and the National Oceanic and Atmospheric Administration (NOAA) created a 3-dimensional (3-D) empirical geoelectric field model. This model provides a map of realistic geoelectric field everywhere in the contiguous United States in quasi-real-time time and is intended to be used as input to power flow software in place of the 1-D geoelectric field calculations, to provide higher accuracy in

calculating GIC. This geoelectric field model has been made possible by enhanced ground-based geomagnetic monitoring at USGS and a growing database of magnetotelluric impedances, a result of an 18-yr long geophysical surveying effort, supported through the National Science Foundation, NASA, and now through the USGS Geomagnetism Program in cooperation with Oregon State University.

A validation of the model was performed in a partnership between the USGS, NOAA, and the Tennessee Valley Authority (TVA) using data from TVA's network of GIC monitors and magnetometers, which provide quasi-real-time reports on GIC data and magnetic field (B-field) data respectively, during significant GMD events.

Using magnetic field time-series data from actual GMD events along with the new 3-D empirical geoelectric field model, NOAA produced a time-varying and geospatially variable geoelectric field. This geoelectric field was then used as input to the system model software using TVA's geographic coordinate data, power system network data and topology, to improve the GIC calculation. The software predicts GIC in each transformer neutral and comparison with data from TVA's GIC monitors allows us to assess the accuracy of this modelling approach.

While many power companies expect only minor effects from GMD storms at the probabilistic intensity recommended by NERC, prudent system vulnerability studies would take into account the geographic variations of geomagnetic field during storm-times and the local effects of the 3-D Earth conductivity, as these may cause more severe effects that challenge operations. Here, we validate a new approach to GIC modeling that accounts for regional variability. This new method can be used in system vulnerability studies to improve grounding models and reduce the likelihood of unpredictable consequences. It can also be used, in conjunction with PowerWorld Simulator or similar tools, to develop system modelling of GICs for real-time operational awareness. This operational information would complement GIC monitoring by providing GIC estimates at all transformer neutrals.

KEYWORDS

Geomagnetically Induced current, GIC, Geomagnetic Disturbance, GMD, Power Transformers, Power systems, Magnetotelluric

INTRODUCTION

TVA grid

The Tennessee Valley Authority (TVA) is a federally-owned, self-financed corporation with the mission to provide navigation, flood control, and electric power in the Tennessee Valley Region. TVA operates the Nation's largest public power system, with a service area covering parts of seven states and 80,000 square miles (over 200 km²) serving 9 million people. It is primarily a wholesaler of power to distributors, but it also sells power directly to larger industrial customers. The TVA transmission system consists of over 16,000 circuit miles (25,000 km) of transmission lines and 513 transmission substations, with transmission operating voltages primarily at 500kV and 161kV. TVA has 52 power transformer banks with high-side windings rated 500kV and configured wye-grounded.

GIC phenomenon

Solar activity can result in disturbances that propagate between the Sun and the Earth and impact Earth's space environment. These events cause elevated electrical currents in Earth's magnetosphere and ionosphere that generate geomagnetic disturbance (GMD). The time-varying magnetic field induces an electric field that drives geomagnetically induced currents (GIC) in the bulk transmission system. For purposes of power-grid studies, GIC change slowly over time and, as such, can be treated as quasi-direct current (dc) in the frequency band 10 μ Hz to 1 Hz. Like other currents, GICs flow according to Kirchhoff's and Ohm's Law. Available paths include extra high voltage (EHV) transmission lines and transformers, and when flowing through power transformer windings, the currents cause part-cycle saturation that can lead to increased reactive power demand, generation of harmonics, and heating.

TVA's SUNBURST network

In cooperation with the Electric Power Research Institute (EPRI), TVA has GIC monitors installed in the 500 kV neutrals of 22 three-phase EHV transformers as well as 12 magnetometers installed across its footprint (Figure 1). The devices communicate data via cell phone links to the contracted support company, which provides quasi-real-time reports during significant events.

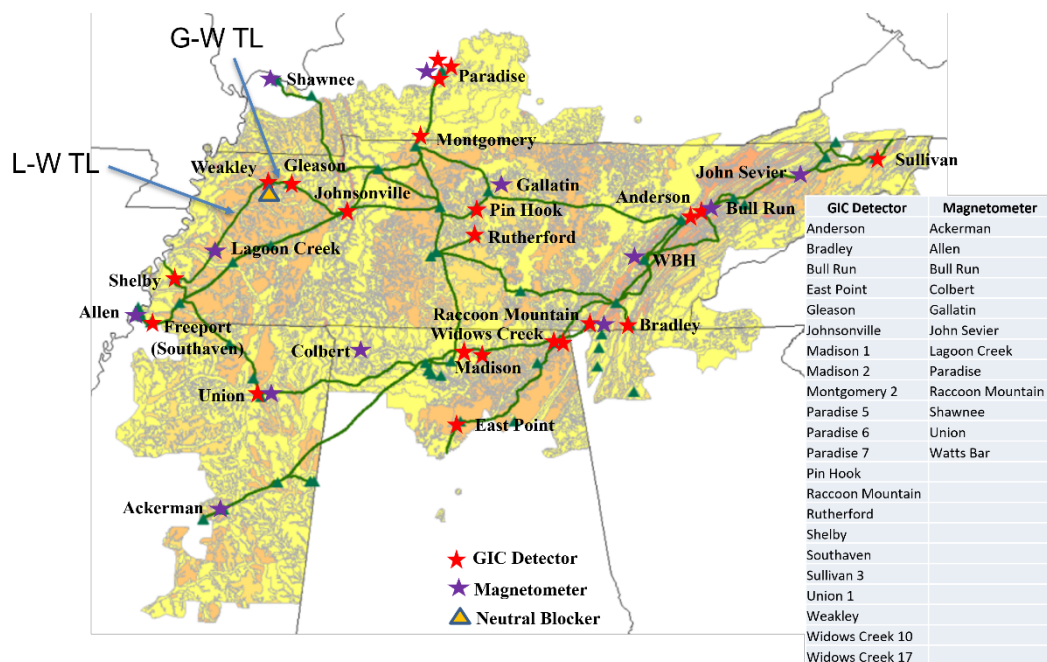


Figure 1. TVA Network of Sunburst Detectors and Magnetometers.

In accordance to the North American Electric Reliability Corporation (NERC) TPL-007-04 standard [1], TVA began developing DC models of its bulk electric system in 2014 in order to assess the impact of GMD on the TVA power system [2] [3]. This work was performed by the Transmission Planning group using planning power flow cases and the PowerWorld Simulator software. In addition, TVA worked with an industry vendor to evaluate their fleet of EHV power transformers to GIC, including magnetic and thermal assessments [4] [5] [6]. Although not required by TPL-007-04, TVA has a keen interest in validating the predicted GIC currents from the models against real-time measurements from GIC monitors. Here, we develop and validate predicted GIC currents based upon a real-time three-dimensional (3-D) model of geoelectric fields, developed by the National Oceanic and Atmospheric Administration (NOAA) and the U.S. Geological Survey (USGS).

This paper describes the efforts made by USGS to obtain empirical magnetotelluric impedances that reflect the Earth's 3-D electrical conductivity structure, NOAA's steps to generate the geoelectric field based on that knowledge, and TVA's process and results of validating the model.

BACKGROUND

Accurate estimation of GICs in the bulk electric system requires a truthful representation of ground-level geoelectric fields everywhere within the footprint of the power grid. Geoelectric fields vary in space and time, and their spatial representation is a function of the Earth's electrical conductivity. Accurate measurements of geoelectric fields, while possible, are extremely challenging and are also contaminated by currents flowing in the electric power lines and by other transient cultural sources. Therefore, making these measurements continuously, consistently, and in multiple locations is impractical. Over the past half century the geophysical method of magnetotellurics (MT) has developed robust MT transfer functions [7], specifically MT impedances, which determine the linear relationship between geomagnetic and geoelectric fields, at a given location. MT surveys can be performed to estimate MT impedances over a given geographical area; once obtained, these frequency-dependent transfer functions can be considered constant on the time scale relevant to human activity and reused indefinitely to obtain geoelectric fields from geomagnetic field measurements.

The geoelectric field calculation in the frequency domain takes the following form:

$$\begin{bmatrix} E_x(\omega) \\ E_y(\omega) \end{bmatrix} = \begin{bmatrix} Z_{xx}(\omega) & Z_{xy}(\omega) \\ Z_{yx}(\omega) & Z_{yy}(\omega) \end{bmatrix} \begin{bmatrix} B_x(\omega) \\ B_y(\omega) \end{bmatrix}, \quad (1)$$

where $B_x(\omega)$, $B_y(\omega)$ are the spectra for the horizontal magnetic field (x – north, y – east), $E_x(\omega)$, $E_y(\omega)$ are the spectra for the surface geoelectric field, and $Z_{ij}(\omega)$ are the complex tensor components of the MT impedance, a quantity that depends on the conductivity of the solid Earth. An inverse Fourier transform can be used to represent the MT impedance as an impulse response in the time domain, and the convolution with the magnetic field can then be performed in real-time to obtain local real-time geoelectric fields,

$$E_h(t) = \hat{\mathbf{Z}} * B_h(t) = \int_0^\infty \hat{Z}(\theta) B_h(t - \theta) d\theta \quad (2)$$

where B_h and E_h are the time-domain horizontal magnetic and electric fields, respectively, and $\hat{\mathbf{Z}}$ is the (real-valued) impulse response. In practice, magnetic field time series B_h are band limited because they are collected as discrete samples after analog filtering through an acquisition system. Therefore, a discrete approximation of equation (2) is used to obtain real-time geoelectric fields [8] [9].

In our study, we use empirical MT impedances obtained using MT survey data. A comprehensive MT survey of the contiguous United States has been performed over the past 18 years by the National Science Foundation (NSF)'s USArray Program [10], continued as the USMTArray [11], funded in separate phases by the U.S. National Aeronautics and Space Administration (NASA) and then the USGS, and is due to be completed in 2024. The NOAA Space Weather Prediction Center and the USGS Geomagnetism Program are using the results to generate geoelectric field maps for the contiguous United States in near real-time.

The model uses geomagnetic observatory data from the USGS Geomagnetism Program and from Natural Resources Canada (NRCan) as inputs to a magnetic field interpolation algorithm [12], which estimates the magnetic field at the MT survey site locations (Figure 2). The MT impedance is convolved with the input magnetic field to obtain the geoelectric field, which is then interpolated to a geographical grid with $\frac{1}{2}$ degree spatial resolution. This geoelectric field best represents our current knowledge of the Earth's 3-D electrical conductivity structure and can be used as input to power grid system models for accurate GIC estimation. As for any GIC estimation model, validation is needed before it can be used industry wide. Here, we present an approach and a case study for NOAA-USGS 3-D Geoelectric Field used as input to an industry-standard PowerWorld Simulation software to estimate GICs in transformer neutrals and validate these estimates against ground measurements.

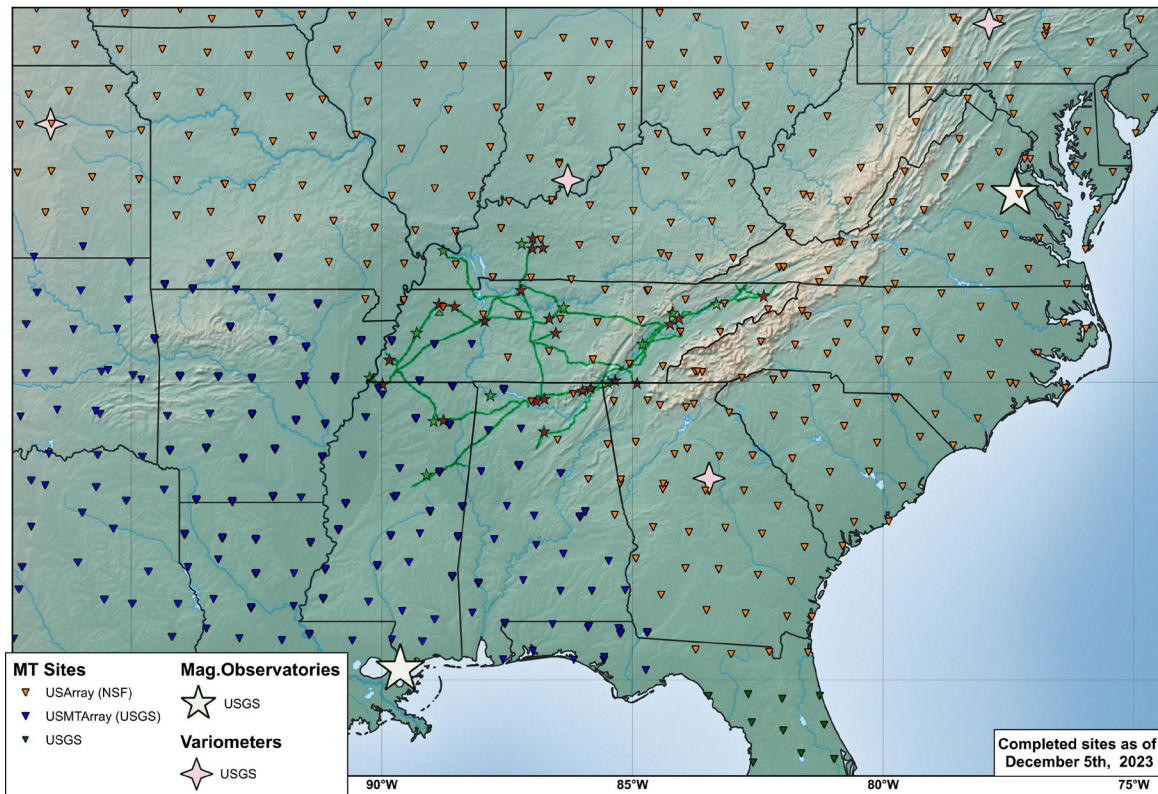


Figure 2. MT impedance locations, the USGS magnetic observatory locations and the newly installed USGS variometer locations shown relative to the TVA distribution network. The USGS variometers were not used in this study but the local TVA magnetometer data (green stars) were assimilated into the geoelectric field model.

For the work of validating a real storm event, a planning case would not accurately depict the system as configured in real-time during the storm. Instead, TVA used a power flow case from their operations State Estimator for the exact time that the storm was at its highest peak. This would model accurate system voltages, any elements that were out of service, system load, system generation, and system transfers that were present during the storm. Using the State estimator presented its own challenges due to modelling discrepancies. In particular, the 3-winding transformers in the State Estimator and planning cases are modelled differently. The first step was to get the transformers to

match the planning case so the GIC data, such as transformer core type, coil resistances, windings, substation latitude and longitude, and grounding resistances, can all be applied to the case. TVA then used the optimized case in PowerWorld Simulator's GIC add on. The GIC add on has an option for Time-Varying Electric Field inputs that uses the modelled geoelectric field as modelled by NOAA/USGS. However, PowerWorld Simulator only computes one time point at a time, so an automation script was written to calculate each time point and output the results to a spreadsheet.

It should also be noted that there are 55 transmission tie lines that connect the TVA system to the surrounding transmission systems. The model does include data from these neighboring systems (lines, transformers, generators), as well as several buses into those systems. Additionally, the DC model data for these external systems are available (i.e., transformer DC winding resistance, substation ground resistance) and were used for NERC TPL-007-04 studies performed by TVA. However, the only DC model data included in the PowerWorld study for this validation effort was for TVA equipment. It is planned to include the DC data of neighboring systems as an enhancement to future studies.

Related, we point out that the TVA system is very extensive, and the analysis was performed for the GIC monitors that are located well within the system. Moreover, in parallel with the PowerWorld simulation, a first-principles approach was used to estimate calculated voltage using a simplified subsystem for the Gleason-Weakley transmission line, whose location allows for this type of analysis. The voltages computed in that manner were nearly identical to those obtained with the full PowerWorld simulation, which indicates that these computations are fairly robust with respect to details of the system extensions at endpoints.

VALIDATION APPROACH

The steps for our validation effort are as follows:

1. Choose a historical GMD storm with meaningful impact to the TVA system, realizing that TVA's southerly latitude tends to reduce impact from GMD storms. For our model validation analysis we analyzed a GMD storm that occurred March 23-24, 2023, having maximum planetary K-index (K_p) of 80, early on the 24th, with three occurrences of K_p 7.
2. Gather GIC and magnetometer data from TVA devices (GIC monitors and magnetometers). TVA provided data from twelve GIC monitors (see Table I), as well as from nine magnetometers, including Paradise, Union, Bull Run, Ackerman, John Sevier, Colbert, Gallatin, Watts Bar, and Lagoon Creek.
3. NOAA creates time-varying E-field using USGS & NRCan operational magnetometer data as well as the TVA magnetometer data, and the 3-D empirical model for Earth conductivity, generating GeoJSON files for each time point, which describe E-field horizontal components for a collection of latitudes and longitudes at $\frac{1}{2}$ degree resolution.
4. TVA performs a study to predict GIC currents using provided input files, focusing on specific sites with meaningful impact. The use of a real-time operation case at the time of the storm is required to ensure the system model topology and load/generation are accurate.
5. Compare predicted GIC with the measured GIC as reported by the TVA GIC monitors.
6. Resolve any discrepancies or issues.

To illustrate the approach used in this analysis, consider two 500 kV transmission lines in the TVA system, the Gleason-Weakley (G-W) 500 kV transmission line along with the Lagoon Creek-Weakley (L-G) 500 kV transmission line. The line routing is illustrated in Figure 1. The predicted geoelectric field components (E_x/E_y , refer to Figure 3 for Gleason-Weakley) in the vicinity of the lines is interpolated and integrated along the line path resulting in a calculated voltage, expressed as a time series v_{GW} , and v_{LW} (refer to Figure 4 for Gleason-Weakley). Using the line resistances, the induced currents in the lines can be derived from the line voltages, and a DC equivalent circuit analysis can be

used to determine the flow of current throughout the system, including the flow through the transformer neutrals in the substations.

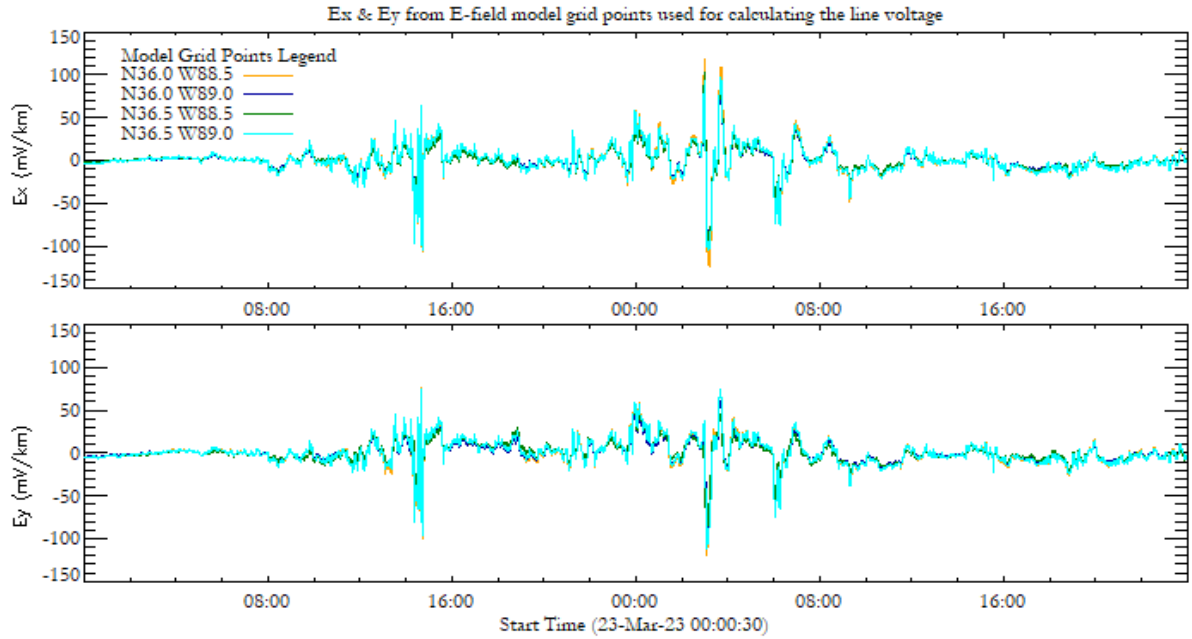


Figure 3. E_x/E_y fields from model grid points for 3/23/2023 near the Gleason-Weakley 500 kV transmission line.

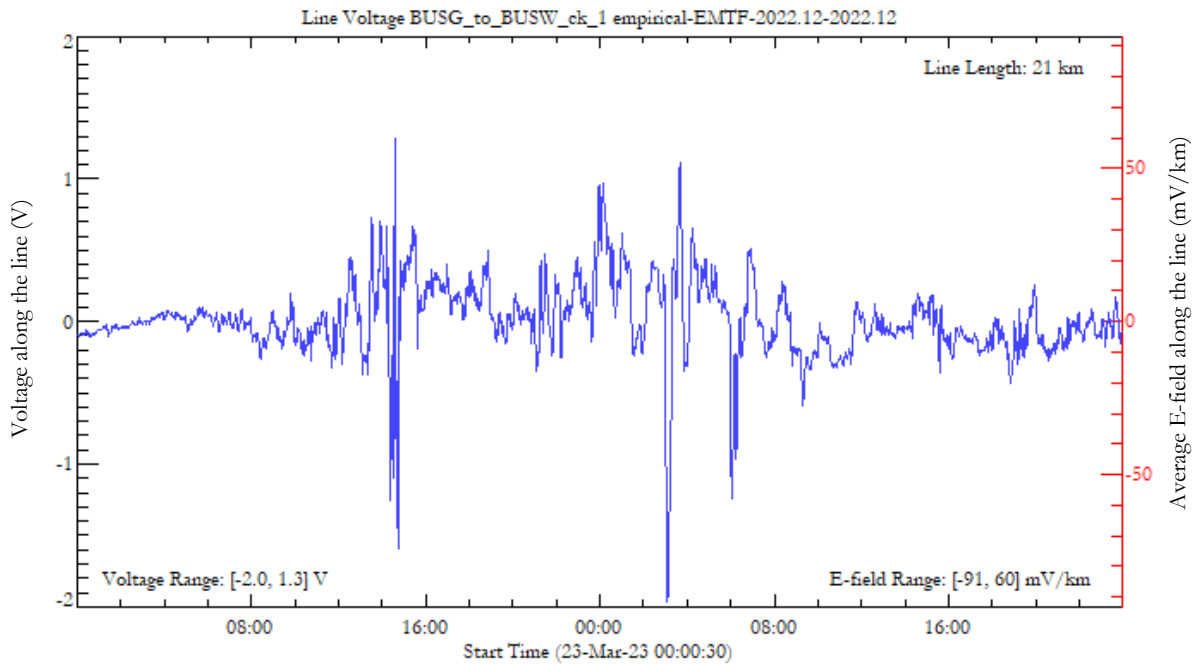


Figure 4. Calculated voltage along the Gleason-Weakley 500 kV transmission line.

RESULTS

The results are provided in the following figures. All are for the GMD storm of March 23-24, 2023. In Figure 5 we compare the GIC time series from the PowerWorld simulation (green line) with the measurements in the Weakley substation. Here we take 1-minute averages of the measurements so as to match the cadence of the geoelectric field model. The time series show good agreement. We note

that the PowerWorld output has been multiplied by -1 because the definition of positive flow in the modelling software is from the system to remote earth, whereas the measurement defines positive to be from remote earth into the system. The modelled and measured values are compared in Figure 6. We find a high correlation of 0.95, and we are able to fit a line (least squares) that has a slope of 0.96, indicating a slight overprediction of the modelled values.

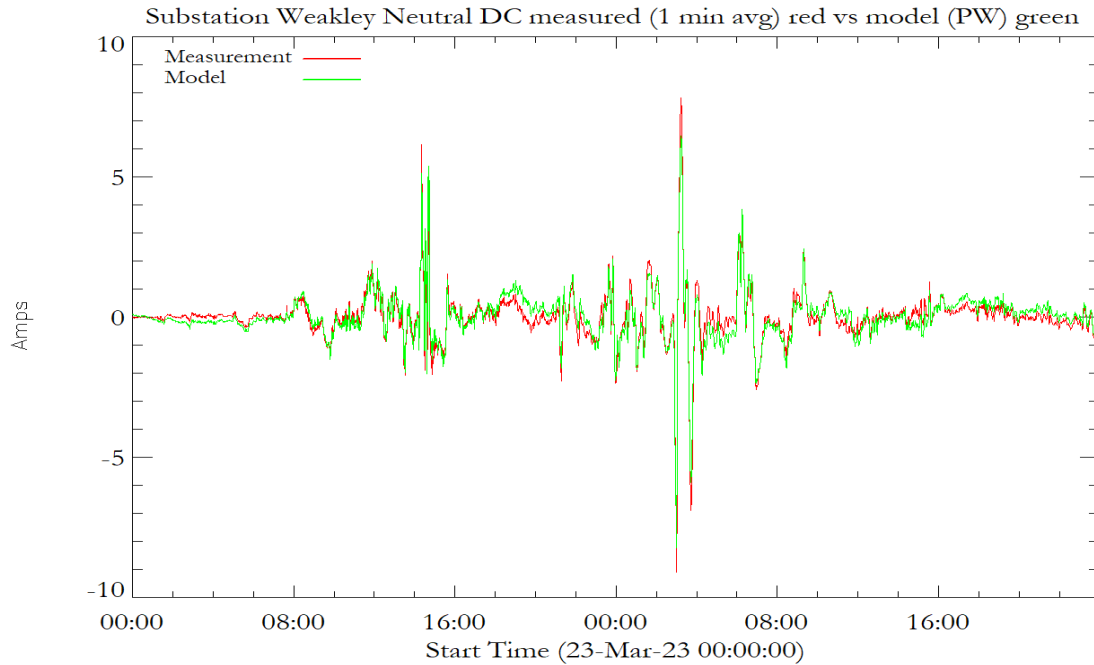


Figure 5. Modelled with PowerWorld (PW; green) and measured (red) DC neutral current at Substation Weakley for March 23, 2023.

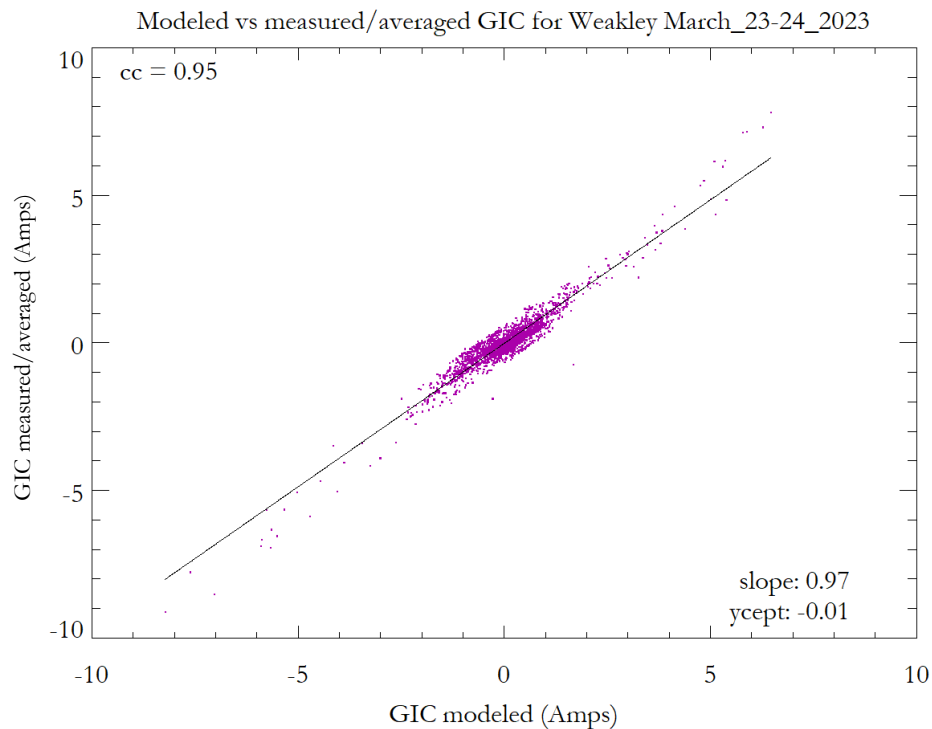


Figure 6. Correlation for GIC in transformer at substation Weakley for March 23, 2023, PowerWorld (model) versus measured.

Similar comparisons were made for 11 other substations, which we summarize, along with the results for Weakley, in Table I (also refer to time-series plots in Figure 7 for Montgomery, Johnsonville, Bull Run, Union). We generally find good correlations until the peak currents drop below 4 Amps, possibly indicating that there are sources of noise in the measurements at and below this level. We include in the table the slope of the best fit line (correlation): values greater than 1 indicate that the model is underpredicting the measurement, and values less than one indicate the model is overpredicting the measurement. Also included in the table is the year when the grounding resistance was last checked for each site, which may motivate the need for carrying out an updated evaluation of the parameter and may be the reason why the scaling is not closer to 1.

Table I. Peak GIC data

GIC Monitor	Peak Current (Amps)	Correlation	Scaling	Sign*	Substation Grounding Resistance (ohms)	Resistance Vintage
Montgomery	20.32	0.96	2.73	-1	0.34	1982
Johnsonville	14.07	0.92	1.22	1	0.13	1958
Weakley	9.66	0.95	0.97	-1	0.18	1976
Bull Run	4.76	0.83	0.87	1	0.09	1964
Union	4.07	0.94	0.58	-1	0.06	1985
Southaven	3.93	0.49	NA	1	0.09	2014
Widows Creek	3.64	0.39	NA	-1	0.08	1965
Shelby	3.54	0.34	NA	1	0.04	1988
Raccoon Mountain	3.5	0.87	1.48	-1	0.28	1979
Rutherford	2.89	0.21	NA	-1	0.17	2010
Bradley	2.53	0.48	NA	-1	1.31	2006
Sullivan	2.27	0.56	NA	1	0.27	1970

*The GIC sensors have generally been installed under the convention that the current flow from the ground into the transformer is positive. The PowerWorld Simulator convention assumes current flow into ground as positive. Where sign="1", the polarity of the modeled GIC was opposite the measured GIC, which is to be expected. It is surmised the GIC sensors may have been inadvertently rolled at sites where sign="1".

DISCUSSION

The comparison of measurements with the calculated GIC in the TVA system as derived from the NOAA/USGS 3-D empirical model and the power system model software shows a high level of correlation in locations where the GIC exceeds 4 Amps, and in some cases we find reasonable agreement with the scaling as well. While further investigation could help to improve the model results for two of the sites (Montgomery and Union), the analysis so far gives confidence that the geoelectric fields are physically realistic and many but perhaps not all parts of the power grid are reasonably well modelled. Once we are able to demonstrate that all parts of the power grid are well modelled, we can build confidence that the results from system vulnerability studies are accurate and that any proposed mitigation actions will be effective.

It is worth noting that many prior GIC calculation methods have relied on estimating linear coefficients directly by fitting a measured GIC to one-dimensional (1-D) models for the geoelectric field components E_x and E_y [13]. Indeed, the linearity of the GICs [14,15] in terms of the local magnetic fields ensures that for any 1-D Earth resistivity model, the GIC can always be represented as a linear combination of the local 1-D geoelectric field components E_x and E_y [16]. However, such relationships are only valid at a location where GIC measurements have been made, and only until the system configuration changes. Instead, we propose a self-consistent validation procedure that does not use GIC measurements to constrain the model but validates any GIC estimates, obtained

independently from an estimated geoelectric field, against such measurements. Then, any modelling errors are independent of the variable system configuration and can be constrained by estimating the uncertainties of the system model and the geoelectric field modelling approach.

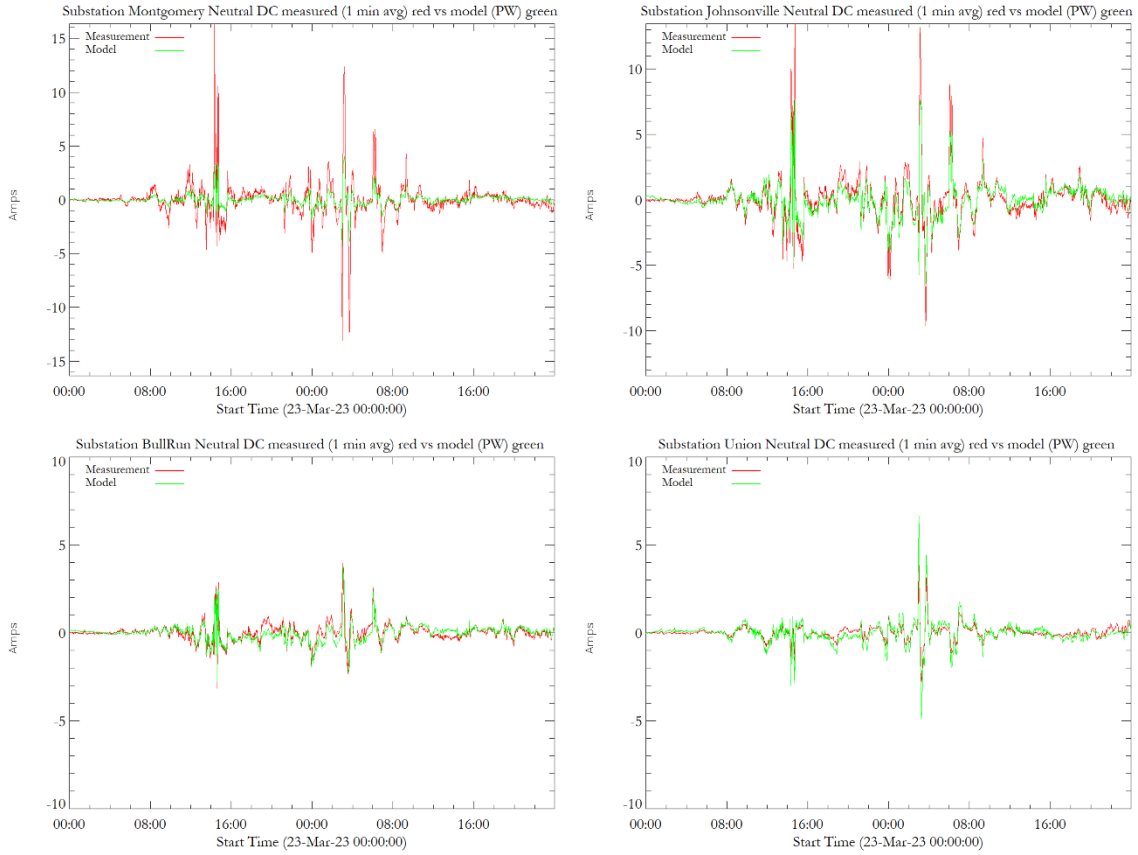


Figure 7. Modelled with PowerWorld (PW; green) and measured (red) DC neutral current at Montgomery, Johnsonville, Bull Run, Union for March 23, 2023.

CONCLUSIONS

Current GMD studies use 1-D Earth resistivity models along with scaled E-field to calculate GIC on the Bulk electric system. The consolidation [17] of MT survey data by USGS in partnership with the NSF USArray program and NASA support have enabled production of a 3-D empirical electric field model. Validation is needed for these models before they are used industry wide.

Here, we develop and validate the first – to the best of our knowledge –end-to-end GIC estimation procedure that uses both the state-of-the-art physics of the geoelectric field estimation in its 3-D representation, through the empirical NOAA-USGS Geoelectric Field Model, and a fully equipped electric power flow software (PowerWorld Simulator) with the best-known representation of active electric power-grid system parameters. We show that in the locations where the grounding and line resistances are best known, this full-physics approach is able to reproduce the measured GICs quite accurately.

We conclude that the correct representation of grounding and line resistances and the system state are critical for accurate GIC estimation. We should specifically note that the uncertainty in the local magnetic field is not a factor in this study because the local TVA magnetometer data have been assimilated into the NOAA-USGS Geoelectric Field Model for this analysis. This was done to eliminate additional sources of error due to the interpolation of the magnetic fields between the sparse

USGS geomagnetic observatories. Other sources of error in this study include the coarse (1 minute) time resolution of the geoelectric field model, compared to the much finer (1 second) time resolution of the GIC measurements; the relatively sparse (70 km) distribution of the empirical MT impedances, and the interpolation of the electric fields to a regular grid; the measurement and estimation errors in the MT impedances; and the conceptual plane-wave approximation of the MT impedances. This latter point refers to the methodology of MT impedance estimation that ensures robust estimates at the expense of elimination of the extremes in the measured magnetic and electric time series. As a result, the MT impedances are not designed to represent storm-time geoelectric fields well. However, as is shown in this study, all these errors compounded still allow for a very accurate representation of GIC currents in the transformer neutrals once the power-grid system parameters are well-known.

Indeed, we have shown that when these parameters are known, and the local magnetometer data are assimilated into the electric field modelling, power-grid system modelling software such as PowerWorld can be successfully used with the NOAA-USGS 3-D geoelectric field outputs for an accurate representation of GICs everywhere in the system. Although the NOAA-USGS geoelectric field outputs are available in near real-time, substantial additional development would need to be performed for real-time GIC estimation using PowerWorld. Specifically, the 3-D geoelectric field inputs to PowerWorld would need to be streamlined, and possibly accepting the streams directly from NOAA would be an option. To obtain the same quality of results, real-time assimilation of local magnetometers or of a dense variometer network would also need to be enabled. With this proof-of-concept, however, such developments are now within reach.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the USGS and NRCan for funding, operating, maintaining, and providing the operational magnetometer data that were used in this study. We thank the funding agencies and investigators who have provided the public their magnetotelluric transfer function products, specifically USArray (NSF), USMTArray (USGS and NASA). We thank Electric Research and Management and the SUNBURST network for providing the GIC measurements. We thank the Finnish Meteorological Institute, Antti Pulkkinen at NASA, and E. Joshua Rigler and Jeffrey Love at USGS for their constructive reviews. This work was supported in part by the NOAA Cooperative Agreement with CIRES, NA17OAR4320101.

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